Hecho con Metabase

Dashboard Case B

Overview

This dashboard presents the experimental evaluation of Case B, a digital complaint workflow implemented in n8n for inter-institutional service orchestration. The workflow coordinates citizen authentication, identity verification, georeferencing, police classification, Fiscalía registration, and institutional notification. The goal of the evaluation is to assess end-to-end latency, interoperability, fault tolerance, and operational traceability under controlled test scenarios.

Experimental Setup

The results shown here were generated with an internal batch runner that sends synthetic complaint requests to the Case B webhook. The workload includes nominal successful executions, business-rule rejections, downstream service rejection scenarios, and latency-stressed requests. This controlled setup supports reproducible evaluation and makes the dashboard suitable for comparative analysis in a research setting.

KPI

Total Executions

total_executions
84

1 fila

True End-to-End Success Rate (%)

true_success_rate_pct
50

1 fila

Business Rejection Rate (%)

business_rejection_rate_pct
26.19

1 fila

Technical Incidents (Dead Letter)

technical_incidents
19

1 fila

Average End-to-End Lantency (ms)

avg_e2e_latency_ms
341.85

1 fila

P95 End-to-End Latency (ms)

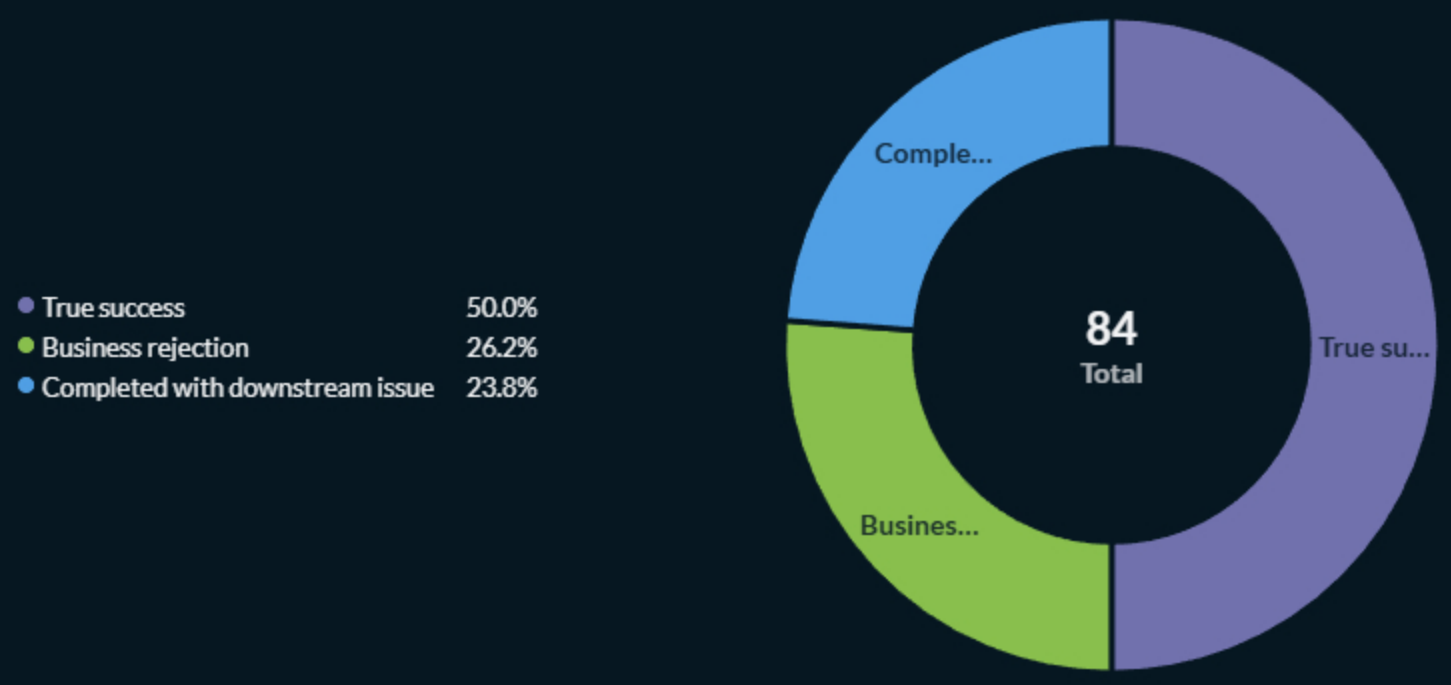
p95_e2e_latency_ms
1,809.8

1 fila

How to read Outcomes

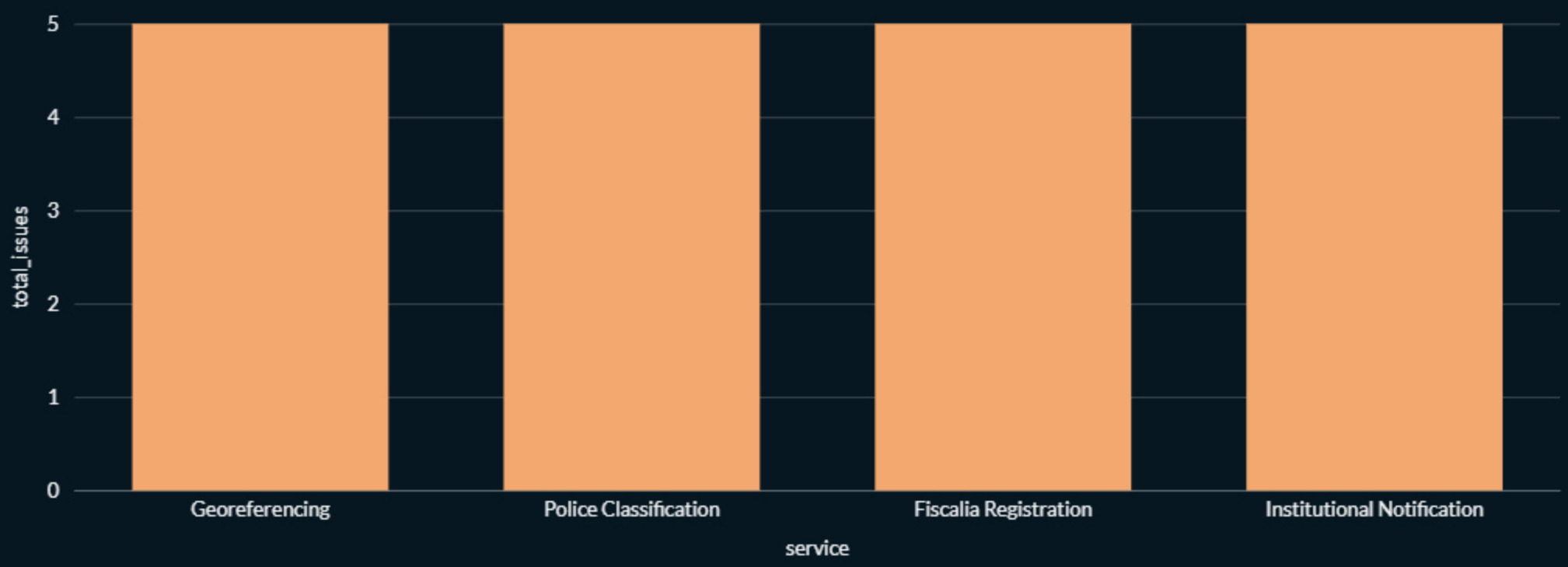
The workflow produces three analytically distinct outcomes. First, true success means that the full process completed and no downstream service reported rejection or failure. Second, business rejection means that the process was validly terminated by domain rules, for example when authentication was denied or identity validation failed. Third, completed with downstream issue means that the workflow reached completion but one or more downstream services reported a rejection or failure state. Technical execution failures are not included in these three categories; they are tracked separately through the dead-letter workflow.

Outcome Composition



Business Rejection Reasons

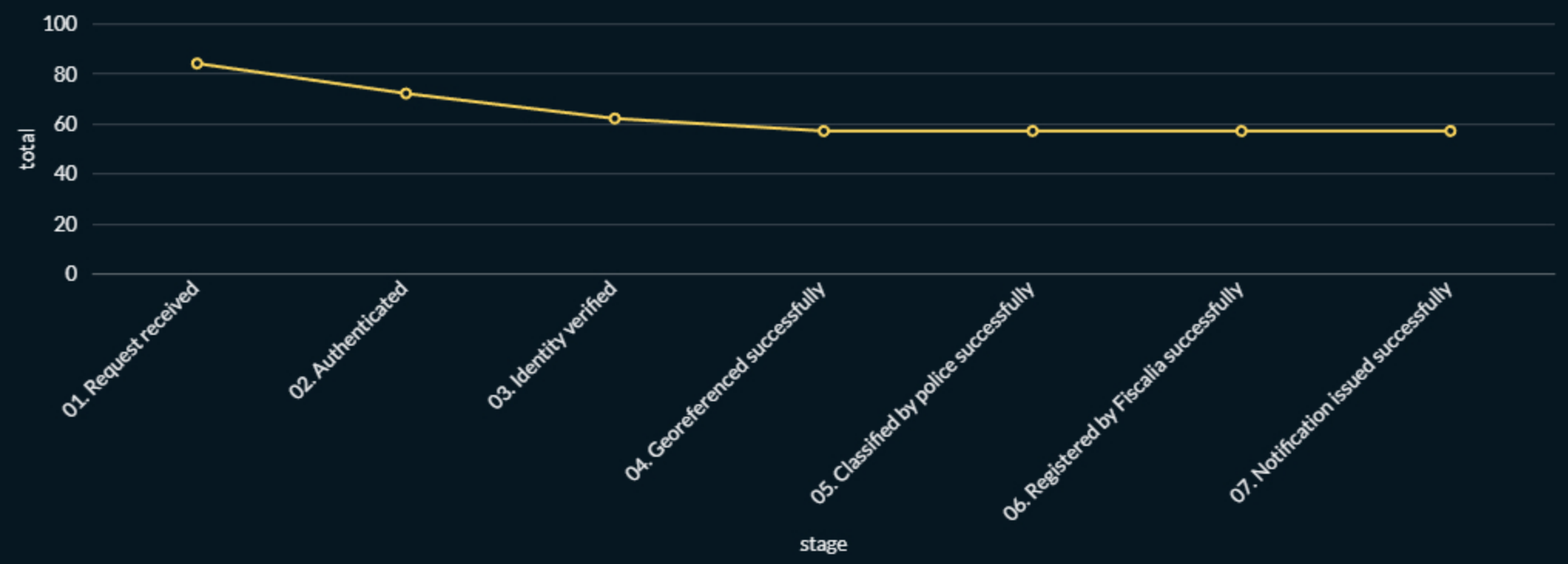
Downstream Issue by Service



Stage Progression

This section shows how requests progress through the major workflow stages. A drop in the early stages typically indicates business-rule rejection, while degradation in the later stages suggests downstream service problems rather than end-to-end execution failure.

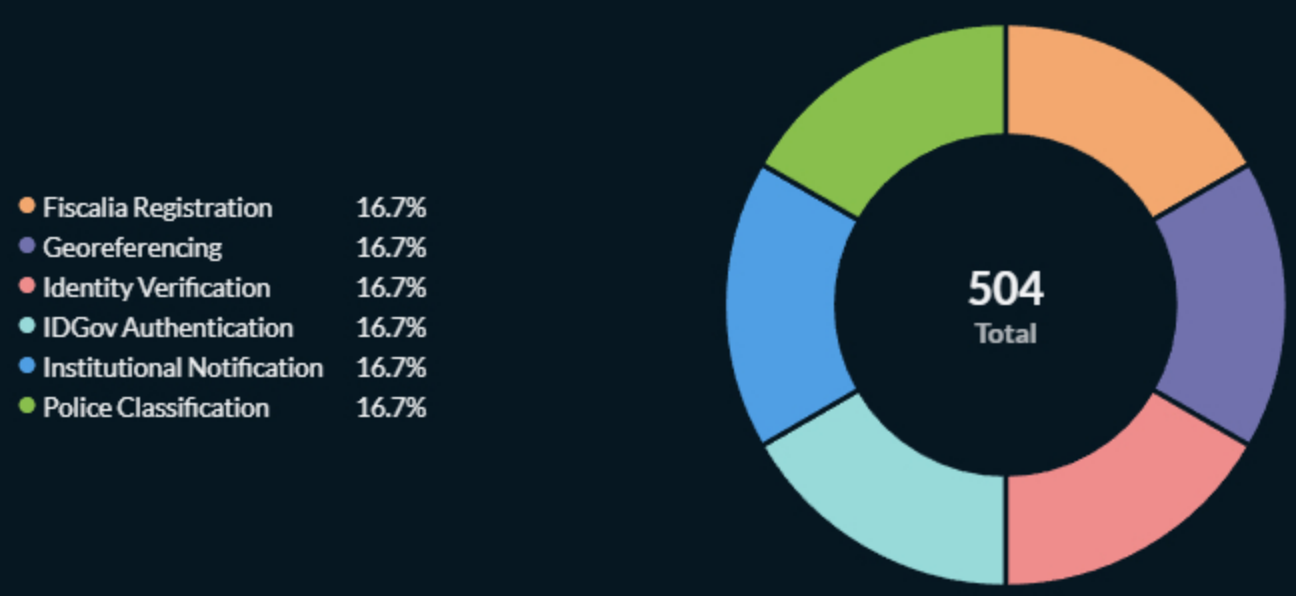
Workflow Funnel



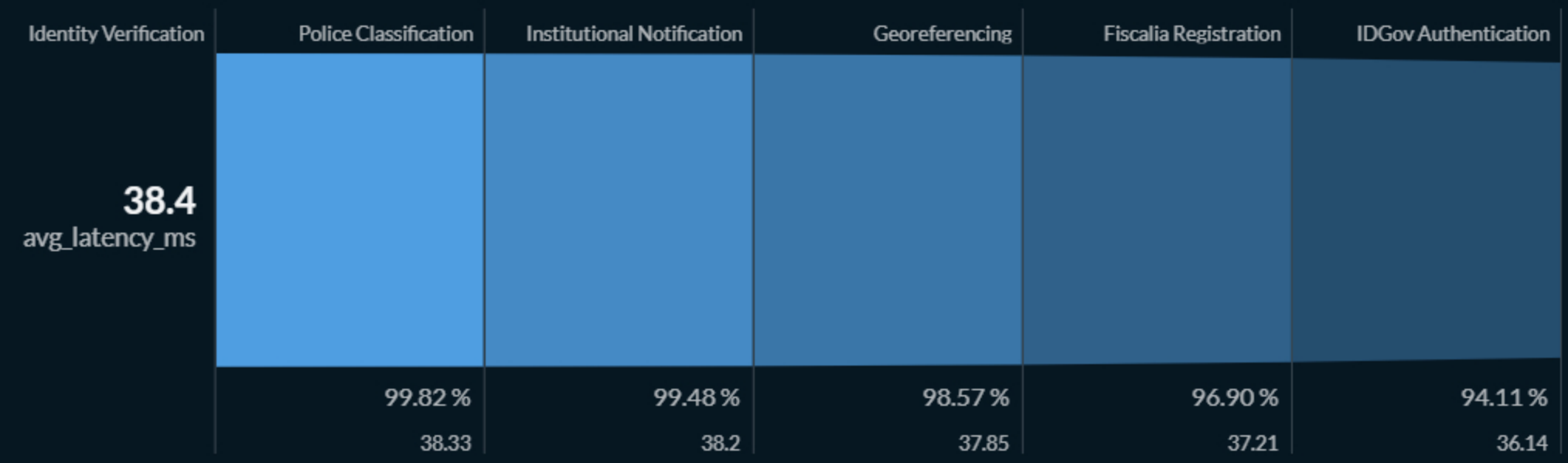
Service Reliability and Interoperability

The complaint workflow depends on six distinct services with different institutional roles. The purpose of this section is to show whether these services behave consistently, whether they become latency bottlenecks, and whether downstream degradation appears even when the full workflow still returns a final response. This distinction is critical for interpreting interoperability in event-driven public service integration.

Service Status Distribution



Average Latency by Service (ms)



P95 Latency by service (ms)

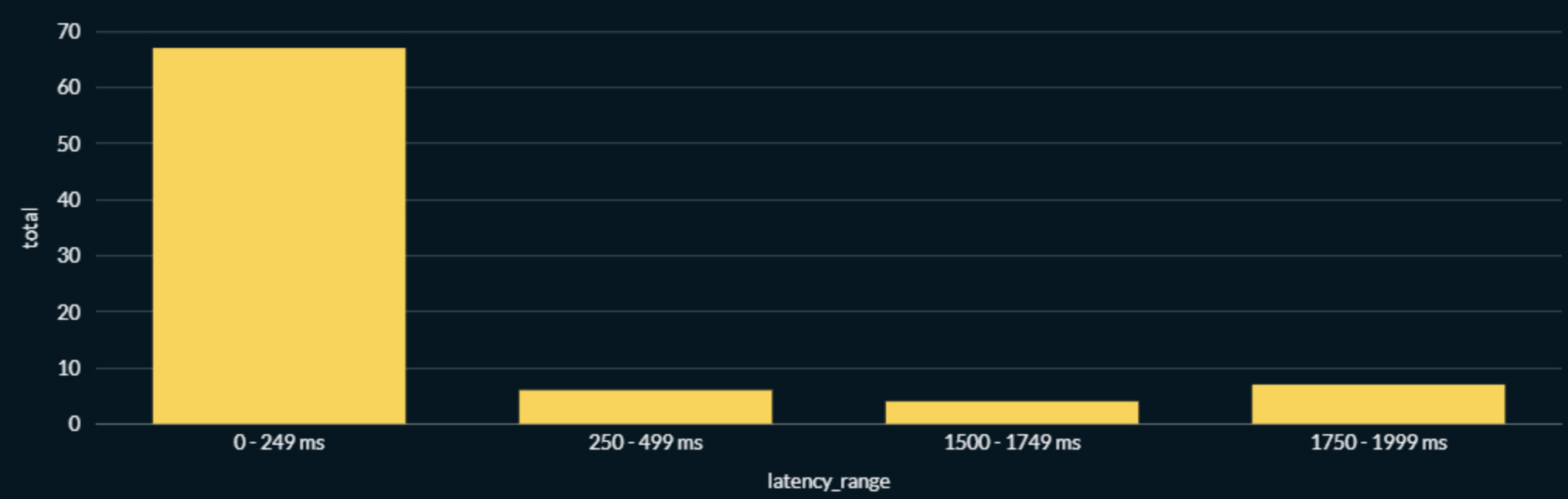
service	p95_latency_ms
Police Classification	304.85
Georeferencing	304.7
Identity Verification	294.85
Institutional Notification	288.1
Fiscalia Registration	284.7

6 filas

Latency interpretation

End-to-end latency is measured from workflow start to final response. Service-level latency is measured independently for each external integration. Together, these indicators show whether the workflow remains stable under nominal execution and whether latency-stressed scenarios create bottlenecks or long-tail behavior.

End-to-End Latency Distribution



Fault Tolerance

Technical execution failures are captured by a dedicated dead-letter workflow. This prevents unexpected runtime errors from being mixed with valid business-rule outcomes. Each dead-letter record stores workflow metadata, execution identifiers, the last executed node, and the associated error payload, enabling post-mortem diagnosis and operational traceability.

Latest Dead-Letter Records

received_at	execution_id	last_node_executed	execution_mode	execution_url
abril 8, 2026, 12:35 AM	507	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/507
abril 8, 2026, 12:35 AM	505	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/505
abril 8, 2026, 12:35 AM	503	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/503
abril 8, 2026, 12:35 AM	501	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/501
abril 8, 2026, 12:35 AM	499	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/499
abril 8, 2026, 12:35 AM	497	HTTP - Autenticacion IDGob	webhook	http://localhost:5678/workflow/SJ6Smavi9Xa4QjEeuU9Qi/executions/497

19 filas

Operational Traceability

The following tables expose execution-level evidence for reproducibility and auditing. They are useful for both engineering review and academic reporting because they make it possible to inspect transaction identifiers, service-level states, fail reasons, latency values, and technical incidents at the level of individual runs.

Latest Workflow Executions

created_at	transaction_id	citizen_id	document_number	incident_category	status	fail_reason
abril 8, 2026, 12:35 AM	tx-caseb-deadletter-georef-001	citizen-dl-georef-001	900000102	GENERAL	ok	
abril 8, 2026, 12:25 AM	tx-caseb-auth-denied-001	citizen-002	1020304051	HURTO	rejected	AUTH_DENIED
abril 8, 2026, 12:25 AM	tx-caseb-auth-denied-001	citizen-002	1020304051	HURTO	rejected	AUTH_DENIED
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-007	citizen-007	1020000006	HURTO	ok	
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-006	citizen-006	1020000005	HURTO	ok	
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-004	citizen-004	1020000003	HURTO	ok	
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-010	citizen-010	1020000009	HURTO	ok	

50 filas

Slowest Executions

created_at	transaction_id	citizen_id	status	fail_reason	latency_ms	service_latency_ms_auth	service_
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-007	citizen-007	ok		1,900	251	
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-004	citizen-004	ok		1,840	254	

abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-001	citizen-001	ok	1,839	279
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-006	citizen-006	ok	1,834	265
abril 8, 2026, 12:12 AM	tx-latency_simulated-1775607148760-009	citizen-009	ok	1,811	269
abril 8, 2026, 12:14 AM	tx-latency_simulated-1775607256696-005	citizen-005	ok	1,803	280

20 filas

Interoperability Reading

Interoperability should not be read only as overall workflow completion. In this case study, a workflow execution may still reach completion even if one of the downstream services reports a rejected or failed state. For this reason, interoperability is evaluated at two levels: end-to-end process completion and service-specific outcome conformance.

Service Outcome Conformance

service	valid_outcomes	total_records	conformance_pct
IDGov Authentication	72	84	85.71
Identity Verification	62	84	73.81
Georeferencing	57	84	67.86
Police Classification	57	84	67.86
Fiscalia Registration	57	84	67.86
Institutional Notification	57	84	67.86

6 filas

Workflow Evidence

The Case B implementation uses six external HTTP integrations, two main decision nodes for early business-rule rejection, three terminal response paths in the main workflow, persistent metric storage in PostgreSQL, and a separate dead-letter workflow for technical failure capture. These design choices support end-to-end observability and make the experiment suitable for technical evaluation in a public-service integration context.

Workflow Evidence Summary

aspect	evidence
Total external integrations	6
Main decision nodes	2
Main terminal response paths	3
Primary metrics table	public.case_b_metrics
Dead-letter table	public.case_b_dead_letter

Primary traceability fields	transaction_id, citizen_id, document_number, status, fail_reason, latency_ms, service_latency_ms_*
Dead-letter traceability fields	workflow_name, execution_id, execution_url, retry_of, last_node_executed, error_message, error_stack, execution_mode

7 filas

Technical Evaluation Summary

The dashboard below is intended not only as an operational monitoring artifact, but also as a presentation-ready research figure. It summarizes whether the workflow satisfies the main evaluation dimensions of the study: end-to-end latency, interoperability across public services, fault tolerance under controlled failure scenarios, and operational traceability for reproducible analysis.

Case B Technical Evaluation Table

category	metric	indicator	result
Technical suitability	End-to-end latency	Average and p95 latency available	Measured through case_b_metrics.latency_ms
Technical suitability	Interoperability	Service-level outcome visibility	Authentication, identity, georeferencing, police, Fiscalia, and notification states reco
Technical suitability	Fault tolerance	Technical incidents routed to dead letter	Captured in case_b_dead_letter
Technical suitability	Traceability	Execution-level identifiers preserved	Transaction, citizen, timestamps, fail reason, and service latency fields stored
Operational sustainability	Reproducibility	Controlled synthetic workload generation	Internal runner supports multiple scenarios and latency-stressed runs
Operational sustainability	Presentation quality	Non-specialist readability	Narrative text cards and staged visual structure

6 filas